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USN-2VX24UE031

Expt.No-2 ¶

Title-Quick Sort

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In [4]: 1 import random
2 import time
3 import matplotlib.pyplot as plt
4
5 def quick_sort(arr):
6     if len(arr)<=1:
7         return arr
8     pivot = arr[len(arr)//2]
9     left = [x for x in arr if x < pivot]
10    middle = [x for x in arr if x == pivot]
11    right = [x for x in arr if x > pivot]
12    return quick_sort(left) + middle + quick_sort(right)
13
14 #input:
15 arr = list(map(int,input("Enter the list of elements separeted by space:").split()))
16 print("Input array:",arr)
17
18 sorted_arr=quick_sort(arr)
19 print("Sorted array:",sorted_arr)
20
21 #measure time taken to sort
22 start_time = time.time()
23 sorted_arr = quick_sort(arr)
24 end_time = time.time()
25 print("Time taken to sort:",end_time - start_time,"second")
26 print("=====")
27
28 # for n>5000
29 n_values = [5000,6000,7000,8000,9000]
30 time_values = []
31
32 for n in n_values:
33     arr = [random.randint(1, 9) for _ in range(n)]
34     start_time = time.time()
35     sorted_arr = quick_sort(arr)
36     end_time = time.time()
37
38     time_taken = end_time - start_time
39     print("time taken to sort",n,"elements:",time_taken,"seconds")
40     time_values.append(time_taken)
41 print("=====")
42
43 # plotting the graph
44 plt.plot(n_values, time_values,color = 'black',marker = 'o')
45 plt.xlabel('Number of elements (n)')
46 plt.ylabel('Time taken (seconds)')
47 plt.title('selection sort time complexity analysis')
48 plt.grid(True)
49 plt.show()
50
51
```

Enter the list of elements separeted by space:2 4 1 7 6

Input array: [2, 4, 1, 7, 6]

Sorted array: [1, 2, 4, 6, 7]

Time taken to sort: 0.0 second

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time taken to sort 5000 elements: 0.0 seconds

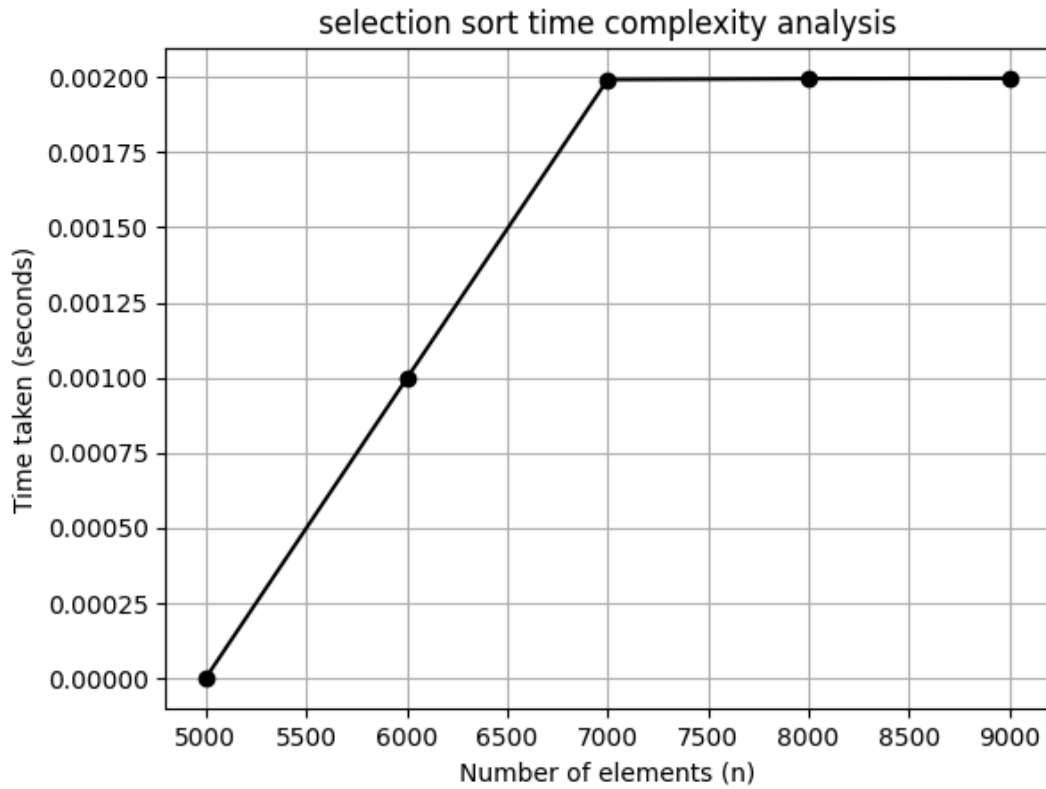
time taken to sort 6000 elements: 0.0009968280792236328 seconds

time taken to sort 7000 elements: 0.0019919872283935547 seconds

time taken to sort 8000 elements: 0.001995086669921875 seconds

time taken to sort 9000 elements: 0.001996278762817383 seconds

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In []:

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